

Table 2. List of samples used in the study.

Sample	Treatment	Description	Contact Angle
REF	none	untreated reference sample	93°
CH	laser texturing	hydrophilic sample with a crosshatch pattern	<1°
CH-H	laser texturing + hydrophobization	hydrophobized sample with a crosshatch pattern	160°
SC	laser texturing	hydrophilic sample with a pattern of shallow channels	<1°
SC-H	laser texturing + hydrophobization	hydrophobized sample with a pattern of shallow channels	154°
DC	laser texturing	hydrophilic sample with a pattern of deep channels	<1°
DC-H	laser texturing + hydrophobization	hydrophobized sample with a pattern of deep channels	154°

2.4. Surface Analysis

The surface of the samples was analyzed using a scanning electron microscope (SEM; Quattro S, ThermoFisher Scientific Inc., Waltham, MA, USA), utilizing both BSE and SE imaging at an accelerating voltage of 10 kV.

Wettability of the samples was evaluated through sessile drop contact angle measurements with a contact angle goniometer (Ossila Ltd., Sheffield, UK). On each surface, 5 drops (approx. 20 μ L) were deposited onto different parts of it and proprietary Ossila software was used to evaluate the resulting images of the contact angle between the water droplet and the surface. All measurements were performed at room temperature.

Figure 1 shows SEM images of all types of surfaces used within the study at different magnifications. On the untreated reference surface REF, only toolmarks from the sandpaper pretreatment are evident. Crosshatch (CH) sample exhibits a distinct cross surface texture, resulting from the treatment in two directions, perpendicular to one another. The direction of the second laser beam pass is more pronounced as that of the first pass due to the relatively high pulse fluence of the laser treatment. The shallow channel (SC) sample exhibits traces of laser texturing which are approximately as wide as the diameter of the laser beam in the focal position. These channels are normally poorly visible and are highlighted through the use of BSE imaging (SE images are shown for all other surfaces). Finally, the bottom row of Figure 1 shows the morphology of the deep channel (DC) sample, where much deeper traces of the laser beam passes are evident and caused by higher pulse fluence. Both the valley and the ridges produced through solidification of molten and ejected material appear porous with cavities in the micron and submicron range. According